Plasma Physics of Beams

USPAS - Beam Physics with Intense Space Charge

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Overview

- We'll use concepts/techniques from plasma physics to describe intense charged particle beams.
- This lecture introduces the *kinetic theory* of plasmas, a self-consistent treatment including both external and self-generated electromagnetic fields.
- We'll derive the Vlasov-Poisson (V-P) equations, which are the basis of the analysis in the rest of the course.
- We start by introducing several useful quantities: the *Debye length*, *plasma parameter*, and *plasma frequency*.
- These quantities are derived for neutral plasmas in thermal equilibrium, but they will also appear in the analysis of charged particle beams (non-neutral plasmas).

Debye length, plasma parameter, and plasma frequency

What is a plasma?

> *A plasma is a gas of charged particles, in which the potential energy of a typical particle due to its nearest neighbor is much smaller than its kinetic energy.* [1]

When this condition is violated, we have a *strongly coupled* plasma. (Example: electrons in metals.)

Debye length

- Consider uniform density of electrons and ions over all space.

$$n_i(\mathbf{x}) = n_e(\mathbf{x}) = n_0 \tag{1}$$

- Add test particle of charge q_T and infinite mass at $\mathbf{x} = 0$.
- How does the test particle change the potential $\phi(\mathbf{x})$?

Add test charge to Poisson equation:

$$\nabla^2\phi(\mathbf{x}) = -\underbrace{\frac{e\left(n_i(\mathbf{x}) - n_e(\mathbf{x})\right)}{\epsilon_0}}_{\text{background}} - \underbrace{\frac{q_T\delta(\mathbf{x})}{\epsilon_0}}_{\text{perturbation}}\ , \tag{2}$$

where e is elementary charge, ϵ_0 is electric constant, $n_e(\mathbf{x})$ is electron density, $n_i(\mathbf{x})$ is ion density, and δ is Dirac delta function.

Electron-electron collisions drive electron distribution to thermal equilibrium at temperature T_e (measured in eV):

$$n_e(\mathbf{x}) = n_0 \exp\left(\frac{e\phi(\mathbf{x})}{T_e}\right) \tag{3}$$

Ion-ion collisions eventually drive ion distribution to equilibrium at temperture T_i.

$$n_i(\mathbf{x}) = n_0 \exp\left(-\frac{e\phi(\mathbf{x})}{T_i}\right) \tag{4}$$

Stop here, before electron-ion collisions relax the entire distribution to equilibrium at the same temperature.

Assume thermal energy dominates: $e\phi \ll T_{i,e}$. Truncate exponential after single term in Taylor series:

$$\exp\left(\frac{e\phi(\mathbf{x})}{T_e}\right) \approx 1 + \frac{e\phi(\mathbf{x})}{T_e} \tag{5}$$

Insert truncated density functions into Eq. (2). Away from the origin, in radial coordinates ($r = |\mathbf{x}|$):

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) = \frac{e^2 n_0}{\epsilon_0}\left(\frac{1}{T_e} + \frac{1}{T_i}\right)\phi(r). \tag{6}$$

Define **Debye length** λ_e and λ_i for electrons and ions:

$$\lambda_e = \sqrt{\frac{\epsilon_0 T_e}{n_0 e^2}}, \qquad \lambda_i = \sqrt{\frac{\epsilon_0 T_i}{n_0 e^2}} \tag{7}$$

And total Debye length $\lambda_D^{-2} = \lambda_e^{-2} + \lambda_i^{-2}$ such that

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) = \lambda_D^{-2}\phi(r). \tag{8}$$

The solution to Eq. (8) is the *Yukawa potential*:

$$\boxed{\phi(r) = \frac{q_T}{4\pi\epsilon_0 r} e^{-r/\lambda_D} = \phi_0(r) e^{-r/\lambda_D}} \tag{9}$$

- Describes *screening* effect: test particle attracts cloud of opposite charge.
- Deybe length λ_D sets length scale for *collective* vs. *collisional* effects.
- Similar potentials appear in nuclear physics.

Plasma parameter

In plasma with average density n_0, average nearest-neighbor distance is $r \sim n_0^{-1/3}$.

Average potential energy between neighboring particles:

$$|q\phi(\mathbf{x})| \sim q^2 r^{-1} \sim q^2 n_0^{1/3}. \tag{10}$$

Typical kinetic energy is much larger than nearest-neighbor potential energy:

$$T \gg q^2 n_0^{1/3}. \tag{11}$$

Therefore,

$$\Lambda \equiv n_0 \lambda_D^3 \gg 1. \tag{12}$$

The **plasma parameter** Λ defines the average number of particles within a *Debye cube* of volume λ_D^3. Eq. (12) is true by definition (weakly coupled plasma).

Plasma frequency

The *plasma frequency* ω_p is derived by considering a one-dimensional slab of uniform density ions and electrons, constrained to the region $0 \leq x \leq L$. Displacing the electrons by a small distance $\Delta \ll L$ will create an electric field $E(\Delta) = -n_0 q \Delta / \epsilon_0$. The resulting harmonic oscillations are described by

$$\ddot{\Delta} + \omega_p^2 \Delta = 0, \tag{13}$$

where

$$\omega_p \equiv \left(\frac{n_0 q^2}{\epsilon_0 m} \right)^{1/2}. \tag{14}$$

The relationship between the plasma frequency and Debye length defines a *thermal velocity* v_p:

$$v_p \equiv \omega_p \lambda_D = \left(\frac{T}{m} \right)^{1/2}. \tag{15}$$

The Debye length λ_D, plasma frequency ω_p, and thermal velocity v_p define the characteristic length and time scales of collective oscillations in the

Accelerator beams are non-neutral plasmas

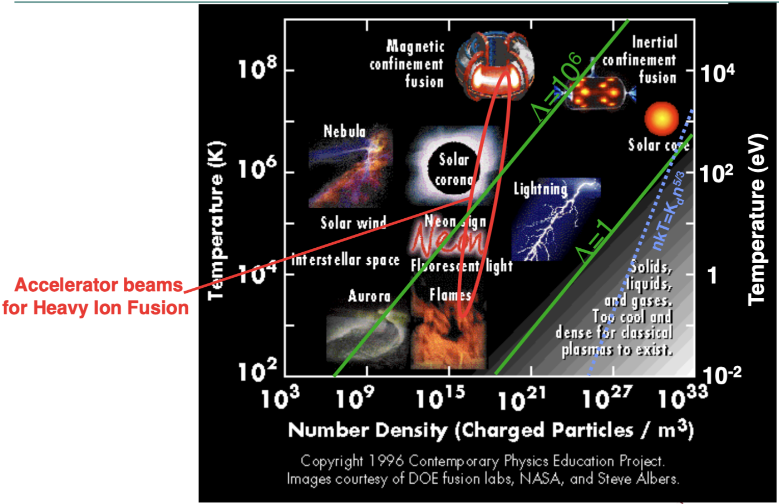

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Figure 1: Plasmas characterized by temperature and number density. Green lines indicate constant plasma parameter Λ. [Updated version?]

Kinetic theory

Klimontovich Equation

- Consider set of N particles moving in six-dimensional phase space defined by the position $\mathbf{x} = (x, y, z)$ and momentum $\mathbf{p} = (p_x, p_y, p_z)$.
- Let $\{\mathbf{X}_i(t), \mathbf{P}_i(t)\}$ be the *orbit* or *trajectory* of particle number i at time t.
- Let $f^m(\mathbf{x}, \mathbf{p}, t)$ be the distribution function:

$$f^m(\mathbf{x}, \mathbf{p}, t) = \sum_{i=1}^{N} \delta\left[\mathbf{x} - \mathbf{X}_i(t)\right] \delta\left[\mathbf{p} - \mathbf{P}_i(t)\right], \tag{16}$$

where δ is the Dirac delta function. Here m stands for *microscopic*, indicating a singular distribution function describing a set of point-particles, rather than a smooth density.

- Total number of particles does not change.

$$N = \int f^m(\mathbf{x}, \mathbf{p}, t) d\mathbf{x} d\mathbf{p}. \tag{17}$$

The charge density $\rho^m(\mathbf{x}, t)$ and current density $\mathbf{J}^m(\mathbf{x}, t)$ are written as

$$\begin{aligned}\rho^m(\mathbf{x}, t) &= \rho^{ext}(\mathbf{x}, t) + q \int f^m(\mathbf{x}, \mathbf{p}, t) d\mathbf{p} \\ &= \rho^{ext}(\mathbf{x}, t) + q \sum_{i=1}^{N} \delta\left[\mathbf{x} - \mathbf{X}_i(t)\right],\end{aligned} \tag{18}$$

$$\begin{aligned}\mathbf{J}^m(\mathbf{x}, t) &= \mathbf{J}^{ext}(\mathbf{x}, t) + q \int \dot{\mathbf{x}} f^m(\mathbf{x}, \mathbf{p}, t) d\mathbf{p} \\ &= \mathbf{J}^{ext}(\mathbf{x}, t) + q \sum_{i=1}^{N} \dot{\mathbf{X}}_i(t) \delta\left[\mathbf{x} - \mathbf{X}_i(t)\right],\end{aligned} \tag{19}$$

where $\rho^{ext}(\mathbf{x}, t)$ and $\mathbf{J}^{ext}(\mathbf{x}, t)$ are from external sources.

The charge and current densities generate an electric field $\mathbf{E}^m(\mathbf{x}, t)$ and magnetic field $\mathbf{B}^m(\mathbf{x}, t)$, which satisfy the Maxwell equations:

$$\begin{aligned}
\nabla \cdot \mathbf{E}^m(\mathbf{x}, t) &= \rho^m(\mathbf{x}, t)/\epsilon_0. \\
\nabla \cdot \mathbf{B}^m(\mathbf{x}, t) &= 0. \\
\nabla \times \mathbf{E}^m(\mathbf{x}, t) &= \partial_t \mathbf{B}^m(\mathbf{x}, t). \\
\nabla \times \mathbf{B}^m(\mathbf{x}, t) &= \mu_0 \mathbf{J}^m(\mathbf{x}, t) + \mu_0 \epsilon_0 \partial_t \mathbf{E}^m(\mathbf{x}, t),
\end{aligned} \tag{20}$$

along with boundary conditions. The microscopic fields and particle trajectories determine the force $\mathbf{F}_i$ on particle i

$$\mathbf{F}_i = \dot{\mathbf{P}}_i(t) = q\left(\mathbf{E}^m\left[\mathbf{X}_i(t), t\right] + \dot{\mathbf{X}}_i(t) \times \mathbf{B}^m\left[\mathbf{X}_i(t), t\right]\right), \tag{21}$$

Note the relationship between the velocity and momentum:

$$\mathbf{P}_i(t) = \gamma_i m \dot{\mathbf{X}}_i(t), \tag{22}$$

$$\gamma_i = \left[1 + \frac{|\mathbf{P}_i|^2}{(mc)^2}\right]^{1/2}. \tag{23}$$

We will study the time-evolution of the microscopic distribution function:

$$
\begin{aligned}
\partial_t f^m(\mathbf{x}, \mathbf{p}, \mathbf{t}) &= \partial_t \left[\sum_{i=1}^{N} \delta\left[\mathbf{x} - \mathbf{X}_i(t)\right] \delta\left[\mathbf{p} - \mathbf{P}_i(t)\right] \right] \\
&= -\sum_{i=1}^{N} \left[\dot{\mathbf{X}}_i(t) \cdot \partial_{\mathbf{x}} + \dot{\mathbf{P}}_i(t) \cdot \partial_{\mathbf{p}}\right] \delta\left[\mathbf{x} - \mathbf{X}_i(t)\right] \delta\left[\mathbf{p} - \mathbf{P}_i(t)\right]
\end{aligned}
\tag{24}
$$

Since $a\delta(a-b) = b\delta(a-b)$, we can replace $\mathbf{X}_i$ and $\mathbf{P}_i$ with $\mathbf{x}$ and $\mathbf{p}$:

$$
\boxed{\left\{\partial_t + \dot{\mathbf{x}} \cdot \partial_{\mathbf{x}} + q\left[\mathbf{E}^m(\mathbf{x}, t) + \dot{\mathbf{x}} \times \mathbf{B}^m(\mathbf{x}, t)\right] \cdot \partial_{\mathbf{p}}\right\} f^m(\mathbf{x}, \mathbf{p}, t) = 0} \tag{25}
$$

[I'm not quite know how to show this. It makes sense if a and b are functions, but in our case we have an operator acting on the delta function.]

Eq. (25) is known as the *Klimontovich Equation*.

Vlasov Equation

The Klimontovich equation, combined with the Maxwell equations, is an exact description of the N-particle system. But this is more information than we need.

Generate smooth distribution function $f(\mathbf{x}, \mathbf{p}, t)$ by averaging $f^m(\mathbf{x}, \mathbf{p}, t)$ over a box of dimensions Δ_x, Δ_p:

$$\begin{aligned} f(\mathbf{x}, \mathbf{p}, t) &= \langle f^m(\mathbf{x}, \mathbf{p}, t) \rangle \\ &= \frac{1}{\Delta_x^3 \Delta_p^3} \int_{\Delta_\mathbf{x}} \int_{\Delta_\mathbf{p}} f^m(\mathbf{x}, \mathbf{p}, t) d\mathbf{x} d\mathbf{p}. \end{aligned} \tag{26}$$

- Assume spatial averaging window much smaller than Debye length: $n^{-1/3} \ll \Delta_x \ll \lambda_D$. (Always possible!)
- Assume momentum averaging window much smaller than thermal momentum: $0 \ll \Delta_p \ll m v_T$.

With the averaging procedure defined, split distribution function into smooth and rapidly fluctuating components:

$$f^m(\mathbf{x}, \mathbf{p}, t) = f(\mathbf{x}, \mathbf{p}, t) + \delta f(\mathbf{x}, \mathbf{p}, t) \tag{27}$$

Averaging procedure gives $\langle \delta f \rangle = 0$.

Do the same to the fields:

$$\begin{aligned} \mathbf{E}^m(\mathbf{x}, t) &= \mathbf{E}(\mathbf{x}, t) + \delta\mathbf{E}(\mathbf{x}, t), \\ \mathbf{B}^m(\mathbf{x}, t) &= \mathbf{B}(\mathbf{x}, t) + \delta\mathbf{B}(\mathbf{x}, t), \end{aligned} \tag{28}$$

where $\langle \delta\mathbf{E} \rangle = \langle \delta\mathbf{B} \rangle = 0$, $\mathbf{E} = \langle \mathbf{E}^m \rangle$, and $\mathbf{B} = \langle \mathbf{B}^m \rangle$.

The course-grained Klimontovich equation gives:

$$\left[\partial_t + \dot{\mathbf{x}} \cdot \partial_{\mathbf{x}} + q\left[\mathbf{E}(\mathbf{x}, t) + \dot{\mathbf{x}} \times \mathbf{B}(\mathbf{x}, t)\right] \cdot \partial_{\mathbf{p}}\right] f(\mathbf{x}, \mathbf{p}, t) = -q\left\langle\left[\delta\mathbf{E}(\mathbf{x}, t) + \dot{\mathbf{x}} \times \delta\mathbf{B}(\mathbf{x}, t)\right]\right\rangle \tag{29}$$

- LHS describes *collective* effects involving *smoothly varying* quantities.
- RHS describes *collisional* effects involving *rapidly varying* quantities.
- Rough estimate: define collisional cross section σ and collision frequency $\omega_c = \sigma n v_t$; ratio of collision frequency to plasma frequency gives approximate relation:

$$\frac{\text{Collisional effects}}{\text{Collective effects}} \sim \frac{1}{\Lambda}. \tag{30}$$

Setting collisional terms to zero in Eq. (29) gives the *Vlasov Equation*:

$$\boxed{\left[\partial_t + \dot{\mathbf{x}} \cdot \partial_{\mathbf{x}} + q\left[\mathbf{E}(\mathbf{x}, t) + \dot{\mathbf{x}} \times \mathbf{B}(\mathbf{x}, t)\right] \cdot \partial_{\mathbf{p}}\right] f(\mathbf{x}, \mathbf{p}, t) = 0} \tag{31}$$

The smoothed fields satisfy the Maxwell equations:

$$\begin{aligned}
\nabla \cdot \mathbf{E}(\mathbf{x}, t) &= \rho(\mathbf{x}, t)/\epsilon_0, \\
\nabla \cdot \mathbf{B}(\mathbf{x}, t) &= 0, \\
\nabla \times \mathbf{E}(\mathbf{x}, t) &= \partial_t \mathbf{B}(\mathbf{x}, t), \\
\nabla \times \mathbf{B}(\mathbf{x}, t) &= \mu_0 \mathbf{J}(\mathbf{x}, t) + \mu_0 \epsilon_0 \partial_t \mathbf{E}(\mathbf{x}, t),
\end{aligned} \tag{32}$$

where $\rho(\mathbf{x}, t)$ and $\mathbf{J}(\mathbf{x}, t)$ are the smoothed charge and current densities:

$$\rho(\mathbf{x}, t) = \rho^0(\mathbf{x}, t) + q \int f(\mathbf{x}, \mathbf{p}, t) d\mathbf{p} \tag{33}$$

$$\mathbf{J}(\mathbf{x}, t) = \mathbf{J}^0(\mathbf{x}, t) + q \int \dot{\mathbf{x}} f(\mathbf{x}, \mathbf{p}, t) d\mathbf{p}. \tag{34}$$

Collisions

[...]

Consequences of Vlasov equation

Liouville Theorem

The Vlasov equation will be the starting point for much of the analysis in the rest of this course. Before proceeding, we will briefly discuss one of its important features: the conservation of phase space density. The Vlasov equation may be viewed as an expression of the *Liouville Theorem*, which states that the phase space density is conserved along particle trajectories (orbits) in Hamiltonian systems:

$$\left.\frac{df}{dt}\right|_{\text{orbit}} = 0. \tag{35}$$

To show this, start from the continuity equation for the phase space density:

$$\partial_t f + \nabla \mathbf{z}(f\dot{\mathbf{z}}) = 0, \tag{36}$$

with $\mathbf{z} = (\mathbf{q}, \mathbf{p})$ containing the canonical coordinate vector $\mathbf{q}$ and momentum $\mathbf{p}$. Given a Hamiltonian $H(\mathbf{q}, \mathbf{p}, t)$, the coordinates evolve according to

$$\begin{aligned} \dot{\mathbf{q}} &= +\partial_{\mathbf{p}} H, \\ \dot{\mathbf{p}} &= -\nabla \mathbf{q} H. \end{aligned} \tag{37}$$

Statistical measures of phase space volume

It will later be useful to describe the phase space density by its low-order moments. The most common quantity used in accelerator physics is the statistical *emittance*. Emittances are calculated from the second-order moments in the covariance matrix Σ:

$$\Sigma = \int \mathbf{z}\mathbf{z}^T f(\mathbf{z}) d\mathbf{z}. \tag{43}$$

For example, with $\mathbf{z} = (x, p_x, y, p_y, z, p_z)$,

$$\Sigma = \begin{bmatrix} \langle xx \rangle & \langle xp_x \rangle & \langle xy \rangle & \langle xp_y \rangle & \langle xz \rangle & \langle xp_z \rangle \\ \langle p_x x \rangle & \langle p_x p_x \rangle & \langle p_x y \rangle & \langle p_x p_y \rangle & \langle p_x z \rangle & \langle p_x p_z \rangle \\ \langle yx \rangle & \langle yp_x \rangle & \langle yy \rangle & \langle yp_y \rangle & \langle yz \rangle & \langle yp_z \rangle \\ \langle p_y x \rangle & \langle p_y p_x \rangle & \langle p_y y \rangle & \langle p_y p_y \rangle & \langle p_y z \rangle & \langle p_y p_z \rangle \\ \langle zx \rangle & \langle zp_x \rangle & \langle zy \rangle & \langle zp_y \rangle & \langle zz \rangle & \langle zp_z \rangle \\ \langle p_z x \rangle & \langle p_z p_x \rangle & \langle p_z y \rangle & \langle p_z p_y \rangle & \langle p_z z \rangle & \langle p_z p_z \rangle \end{bmatrix}. \tag{44}$$

For a phase space of dimension K, the K-dimensional emittance ε is the square root of the determinant of the covariance matrix.

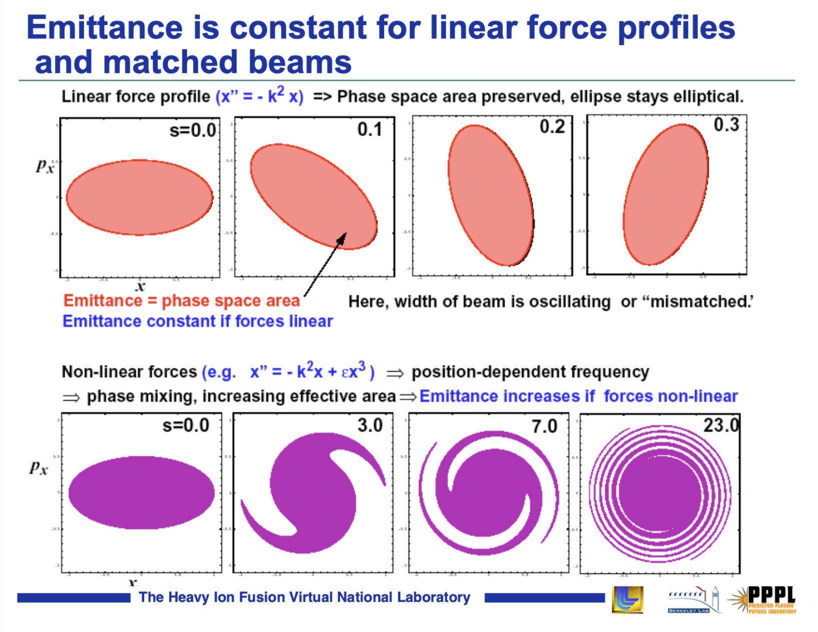

Figure 2: Root-mean-square (rms) emittance growth under nonlinear forces.